

Doerun Georgia Temperature Record Diagnostic Report

Date of Analysis: 2026-09-16

Source: Wundeground KGADOERU5

Data Description: Median temperature values aggregated monthly.

Variable Analyzed: Median temperature

Units: degrees fahrenheit

Observation frequency: Monthly

Observation count: 80

Time Range: 2020-01-31 to 2026-09-30

1. Summary and Overview

1.1 Executive Summary

This report presents a structured diagnostic evaluation of temperature behavior using time-series analytical techniques. The dataset was processed and evaluated for trend structure, distributional behavior, autocorrelation, cyclic characteristics, and anomaly presence. Results indicate no statistically detectable linear trend at the 0.05 level, meaningful persistence near a lag of 12 months, and a possible recurring spectral period near 11.57 months. Distribution testing suggested the Beta distribution provided the strongest distributional fit.

1.2 Structural Summary Panel

Diagnostic Area	Structural Result
Sample Size	80 observations
Trend	Limited evidence of linear trend
Distribution Shape	Low skew
Distribution Fit	Beta provided the only acceptable fit
Outliers	None detected
Autocorrelation	Moderate autocorrelation near a lag of 12 months
Spectral Structure	Possible recurring pattern near 11.57 months
Rolling Variability Structure	Rolling standard deviation is nearly constant throughout the series.

1.3 Overview

This report provides a structured diagnostic analysis of the selected time series, median temperature. The goal is to identify statistical structure, trend behavior, distribution characteristics, and possible readiness for deeper analysis.

1.4 Analyst Context Note

This analysis focuses on understanding the statistical structure of the dataset, including variability, trend behavior, distribution characteristics, and possible cyclic structure. Results should be interpreted as exploratory diagnostics intended to support further investigation rather than definitive causal conclusions.

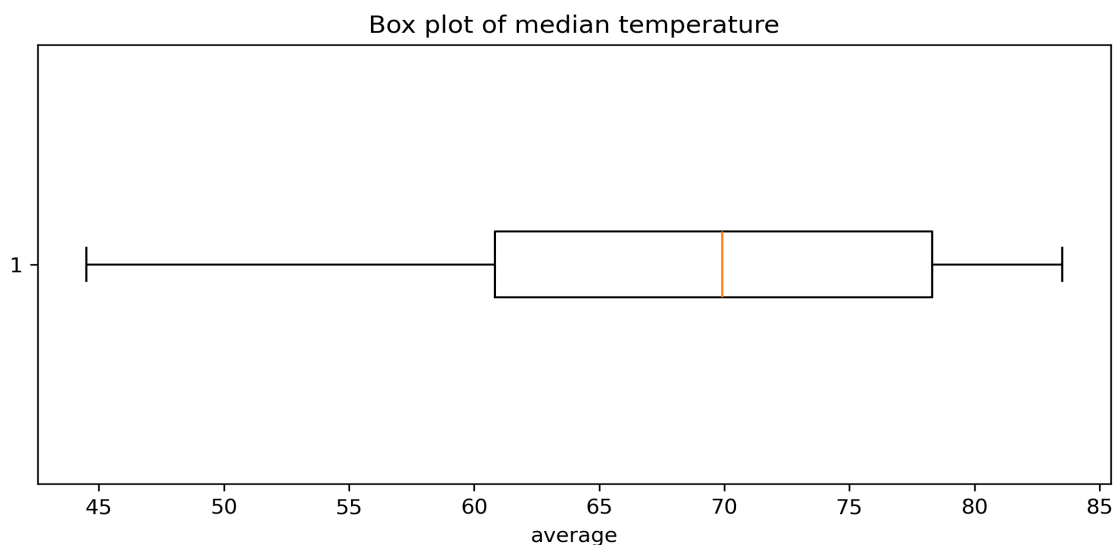
2. Summary Statistics

2.1 Summary Statistics table

Statistic	Value
Count	80
Mean	68.79
Median	69.93
Std	11.02
Variance	121.42
Min	44.50
Q1	60.84
Q3	78.30
Max	83.50
Skew	-0.42
Kurtosis	-1.04
Lower Outlier Threshold	34.64
Upper Outlier Threshold	104.49
Number of Lower Bound Outliers	0
Number of Upper Bound Outliers	0

2.2 Box-and-Whisker plot

A box-and-whisker plot is shown below to visualize the distribution and outliers in the median temperature dataset. Note, multiple outliers may overlap visually in the box plot due to identical or near identical x-positioning.



3. Trend & Residual Analysis

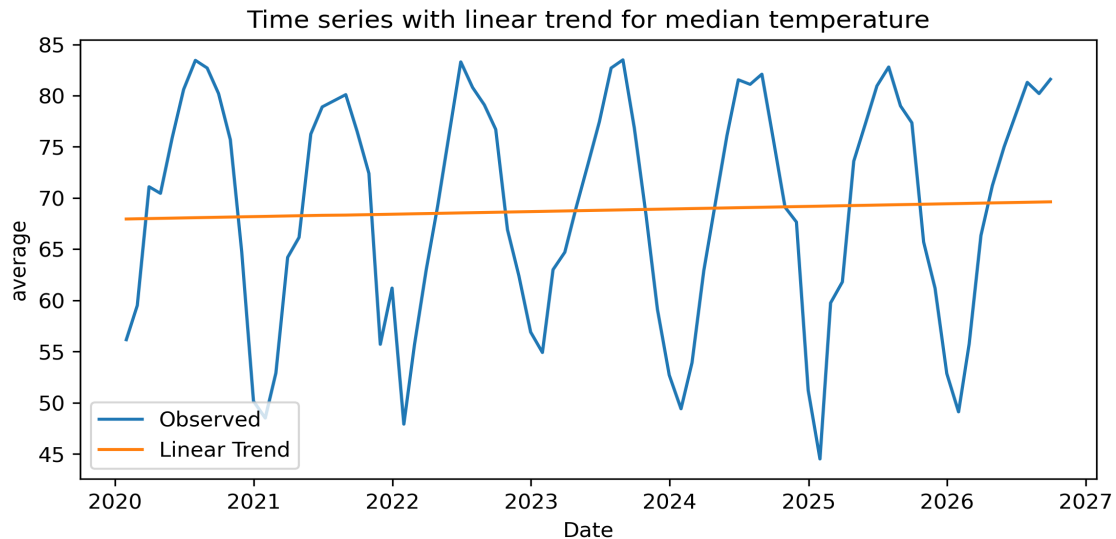
3.1 Linear Trend

Slope: 0.021241

R-squared: 0.0020

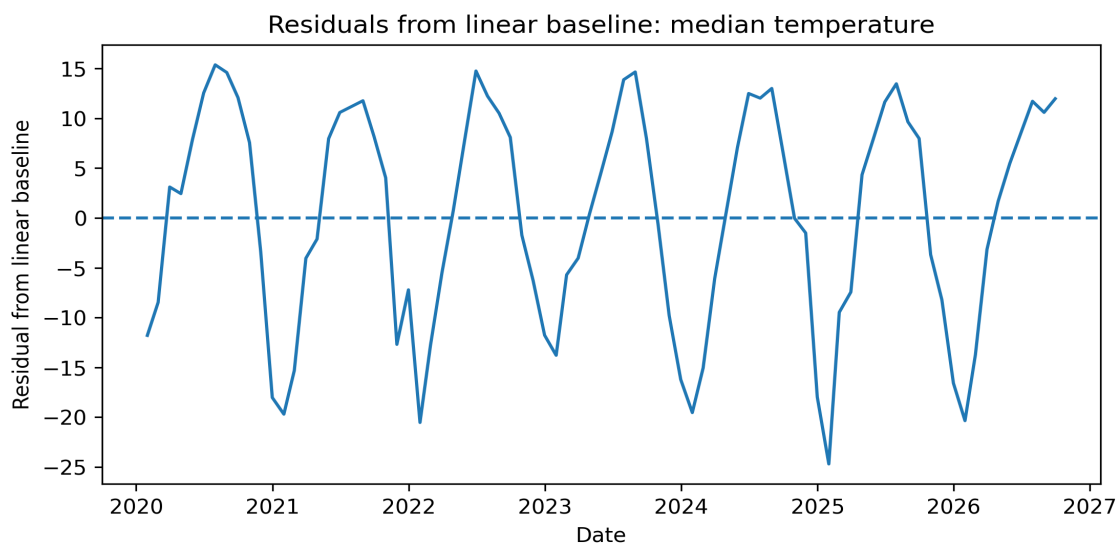
Trend p-value: 0.6932

No statistically significant linear trend was detected at the 0.05 level for the median temperature.



3.2 Linear Residuals

Because the fitted linear trend explains little of the variation, this plot should be interpreted as a baseline-deviation view.



4. Distribution Comparison

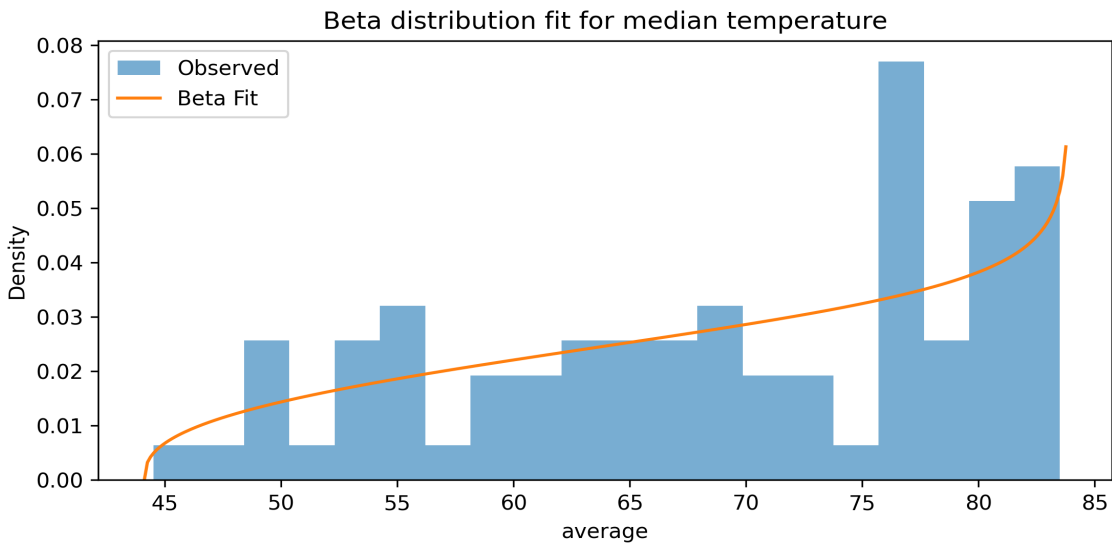
4.1 Beta distribution fit

A Beta distribution was fitted to the median temperature using a bounded support interval.

Metric	Value
Alpha	1.3883
Beta	0.8717
Bound Mode	Empirical
Lower Bound	44.1100
Upper Bound	83.8900
KS Statistic	0.0859
KS p-value	0.5665

The KS test does not reject the Beta distribution fit at the 0.05 level.

The Beta model used an empirical support interval derived from the observed range with a small amount of boundary padding.

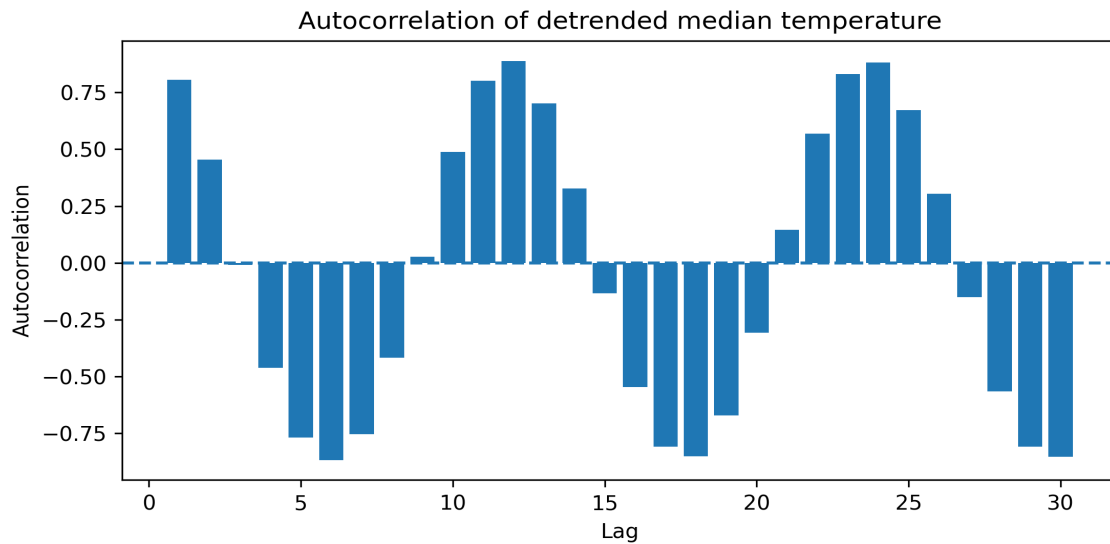


5. Lag and Periodicity

5.1 Autocorrelation Analysis

Autocorrelation measures whether later values in the dataset are similar to earlier values of itself.

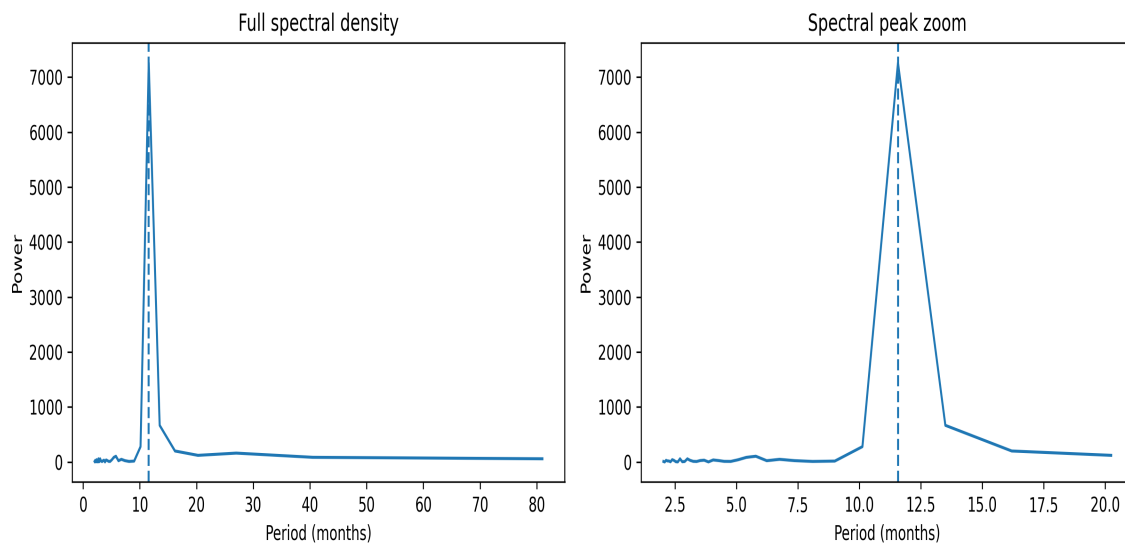
The strongest observed autocorrelation occurred near a lag of 12 months for the median temperature, with a correlation of 0.8897.



5.2 Spectral Analysis

Spectral analysis estimates which periods contribute the most power to the detrended time series. This helps identify possible recurring cycles.

The strongest peridogram peak detected occurs at approximately 11.57 months for the median temperature.



6. Local Variability & Extreme Values

This section evaluates local variability changes in the series.

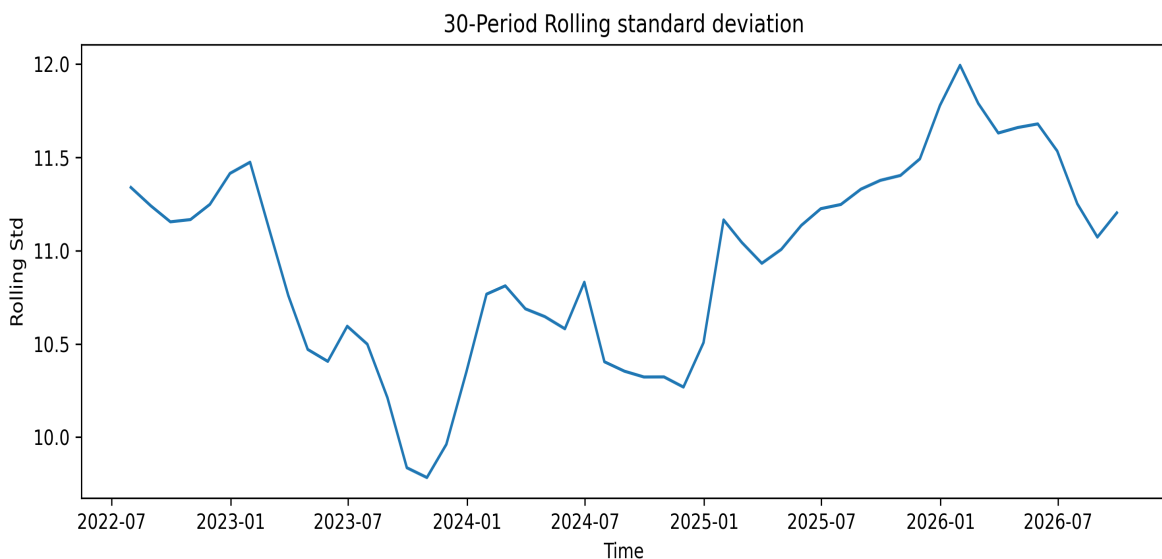
6.1 Rolling Variability

Rolling Standard Deviation Average: 10.95

Rolling Standard Deviation Maximum: 11.99

Rolling Variance Average: 120.21

Rolling Variance Maximum: 143.86



The average rolling standard deviation was 10.95, while the maximum rolling standard deviation reached 11.99.

The average rolling variance was 120.21, while the maximum rolling variance reached 143.86.

6.2 Relative Extreme Values

Threshold	Value	Count Above	Rate Above
90th Percentile	81.55	8	10.0%
95th Percentile	82.70	4	5.0%
99th Percentile	83.46	1	1.2%

Percentile exceedance rates provide a threshold-based view of extreme values. Higher exceedance counts indicate how often the series entered elevated-value zones relative to its own historical distribution. These thresholds do not define operational danger by themselves, but they provide a useful starting point for monitoring unusually high values.

6. Results

- No IQR-based outliers for median temperature were identified.
- The analysis did not support a statistically significant linear trend for the median temperature.
- The Beta distribution provided the only statistically acceptable fit.
- Meaningful autocorrelation was noted around a lag of 12 months.
- The strongest periodogram peak found in the data lies around 11.57 months.
- Local variability is similar throughout the time series.